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EEG-based emotion recognition: Review of commercial EEG devices and machine learning techniques

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ABSTRACT

Emotion recognition based on electroencephalography (EEG) signal features is now one of the booming big data research areas. As the number of commercial EEG devices in the current market increases, there is a need to understand current trends and provide researchers and young practitioners with insights into future investigations of emotion recognition systems. This paper aims to evaluate popular consumer-grade EEG devices' status and review relevant studies that examined the reliability of these low-cost devices for emotion recognition over the last five years. Additionally, a comparison with research-grade devices is conducted. This paper also highlights EEG-based emotion recognition research's key areas, including different feature extraction capabilities, characteristics, and machine learning algorithms. Finally, the main challenges for building an EEG-based emotion recognition system, focusing on the data collection process with commercial EEG devices and machine learning algorithms' performance, are presented.

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1. Introduction

In the past decades, electroencephalography-based emotion recognition studies have gained popularity in many disciplines. However, available scientific data about emotional states and their structure is still limited. A successful exploration of emotions based on electroencephalography (EEG) has gained significant interest in medicine and various fields such as computer science, marketing, education, and entertainment. The following Table 1 shows a list of recent studies that focus on EEG-based emotion recognition. Popular databases are used for the literature search, such as Google Scholar, Springer Link Online Libraries, Multidisciplinary Digital Publishing Institute (MDPI) and Science Direct (Elsevier), with main keywords as electroencephalography, EEG, emotion classification, emotion prediction and emotion classification. In EEG, electrodes are placed on a patient's scalp, primarily to detect neurons' electrical activity in the cerebral cortex (Sheehy, 1984). EEG does not record the activity of a single neuron but rather detects the signals created when populations of neurons are active at the same time. It records signals from small areas of the brain surrounding each electrode. EEG provides an image of electrical activity in the brain represented as waves of varying frequencies, amplitude and shape. It can be used to measure brain activity during an event like the completion of a task or the observation of a stimulus or to measure spontaneous brain activity in the absence of a specific event.

Despite the potential applications in different areas, the utilization of EEG technology beyond the laboratory is limited by enormous time spent for equipment and software setup, user immobility caused by a bunch of connection wires, and high cost of research-grade devices (Grummett et al., 2015). Recently, engineers and EEG researchers have explored ways to overcome these problems and enable smooth and reliable EEG signal collection in a real-life environment (Callan et al., 2015; Ratti et al., 2017). Several types of low-cost consumer-grade EEG devices with wireless signal transmission and gel-free EEG data collection are available in the market today. Such design makes EEG devices more portable and easy to use in a daily life environment. Although the research-grade EEG devices remain preferable for emotion recognition in clinical institutions, similar studies have been conducted using consumer-grade EEG devices and reliable results have been presented (Maskeliunas et al., 2016).

With EEG, brain activity signals can be captured and investigated via machine learning algorithms to predict emotion recognition patterns and convert them into commands. Commonly either a classification or regression model is employed for this task. The type of algorithm is determined by the outputs and the stimuli presented during the experiment. Thus, when the emotions are depicted as a multi-dimensional scale, and each scale is a continuous variable, a regression would be preferred. However, when emotions are reflected as a separate category or emotional scales are divided into subcategories, a classification algorithm would be applied. This study aims to conduct a comprehensive investigation of the status

of low-cost consumer-grade EEG devices used to evaluate emotional states and provide insight into the quality, capabilities, characteristics, and machine learning classification methods used for these EEG devices. Furthermore, research challenges are discussed, focusing on the data collection process with affordable EEG devices and machine learning algorithms' performance.

The rest of the paper is organized as follows: Section 2 covers the review of different types of affordable EEG devices; section 3 presents EEG characteristics, including a review of previous studies and the methodologies used; section 4 presents discussions on reviewed studies; challenges with consumer-grade EEG devices are presented in Sections 5; Section 6 is the conclusion.

2. Consumer-grade EEG devices

Over the last ten years, the purpose of non-invasive EEG devices has transformed from research-based applications into more commercial use (Rashid et al., 2018; Ratti et al., 2017; Wei and Ma, 2017). Purchasing research-grade EEG equipment can be costly, which can reduce research output and cutoff research funding. Many researchers are sourcing for a cheaper option of an EEG device that produces comparable data quality. The current market offers various consumer-grade EEG devices with low cost, simple assembly/disassembly, and satisfactory performance. The main concern would be the validation of these products considering the quality and reliability of the obtained EEG data. Six low-cost consumer-grade EEG devices are presented in this study. Namely, Emotiv EPOC X, Emotiv EPOC+, Emotiv INSIGHT, Emotiv EPOC FLEX, OpenBCI, and NeuroSky MindWave. A summary of these six affordable EEG devices, including the year of release, price, number of electrodes, and sampling capacity, are presented in Table 2.

Out of the six devices, EPOC FLEX is the most expensive model as it is designed mainly for medical applications, with a starting price of \$1699. The rest of the Emotiv products range from \$299–\$799, depending on sensors count and specifications. Emotiv offers three licensing plans for a variety of needs. There are a free license and a pro license. The free license is designed for brain-computer interface APIs. Pro license is designed for neuroscience research and includes student, academic, and business types of plans. Pro license enables viewing and exporting raw EEG and performance metrics data in real-time. The business license is designed for commercial use involving research and product development. Emotiv products are wireless, giving the possibility of more free movement to the respondent. The OpenBCI stands for an open-source brain-computer interface. The OpenBCI products can be purchased in the range of \$750–\$1080, depending on the number of sensors. The OpenBCI also offers a “print-it-yourself” option, enabling a user to 3D print the headset, which is also compatible with conventional head caps. The manufacturer is committed to open-access and budget-friendly EEG solutions, offering a broadened choice of brain studies tools. The NeuroSky MindWave model has

Table 1
Recent studies concerning emotion recognition measuring EEG data.

Reference	Title
(Asghar et al., 2019)	EEG-Based Multi-Modal Emotion Recognition using Bag of Deep Features: An Optimal Feature Selection Approach
(Huang et al., 2019)	An EEG-Based Brain-Computer Interface for Emotion Recognition and Its Application in Patients with Disorder of Consciousness
(Jingru et al., 2019)	Experimental Research on Emotion Recognition Based on Brain-Computer Interface and Brain Waves
(Krishna et al., 2019)	Emotion classification using EEG signals based on tunable-Q wavelet transform
(Lakhan et al., 2019)	Consumer Grade Brain Sensing for Emotion Recognition
(Sorinasa et al., 2019)	A new model for the implementation of positive and negative emotion recognition
(Zeng et al., 2019)	EEG Emotion Classification Using an Improved SincNet-Based Deep Learning Model
(Alazrai et al., 2018)	EEG-Based Emotion Recognition Using Quadratic Time-Frequency Distribution
(Ganesh et al., 2018)	Participant-dependent and Participant-independent Classification of Emotions using EEG Signals
(Katsigiannis and Ramzan, 2018)	DREAMER: A Database for Emotion Recognition through EEG and ECG Signals from Wireless Low-cost Off-the-Shelf Devices
(Liu et al., 2018)	Real-Time Movie-Induced Discrete Emotion Recognition from EEG Signals
(Zhuang et al., 2018)	Investigating Patterns for Self-Induced Emotion Recognition from EEG Signals
(Chai et al., 2017)	A Fast, Efficient Domain Adaptation Technique for Cross-Domain Electroencephalography(EEG)-Based Emotion Recognition
(Al-Galal et al., 2016)	EEG-Based Emotion Recognition while Listening to Quran Recitation Compared with Relaxing Music Using Arousal-Valence Model
(Kumar et al., 2016)	Bispectral Analysis of EEG for Emotion Recognition
(Zhang et al., 2016)	Relief-Based EEG Sensor Selection Methods for Emotion Recognition
(Zheng et al., 2016)	Unobtrusive and Multi-modal Wearable Sensing to Quantify Anxiety
(Ahmad et al., 2015)	Discriminating the Different Human Brain States with EEG Signals using Fractal Dimension: A Nonlinear Approach
(Javaid et al., 2015)	Real-Time EEG-based Human Emotion Recognition
(Thammasan et al., 2015)	EEG-based Investigation of Music Familiarity and Emotion
(Liu and Sourina, 2014)	EEG-based Subject-Dependent Emotion Recognition Algorithm Using Fractal Dimension
(Peng et al., 2014)	EEG-based emotion recognition with manifold regularized extreme learning machine
(Wang et al., 2014)	Emotional state classification from EEG data using machine learning approach
(Zheng et al., 2014)	EEG-based emotion classification using deep belief networks
(Koelstra et al., 2012)	DEAP: A Database for Emotion Analysis using Physiological Signals
(Nie et al., 2011)	EEG-based Emotion Recognition during Watching Movies
(Inventado et al., 2010)	Predicting Student's Appraisal of Feedback in an ITS Using Previous Affective States and Continuous Affect Labels from EEG Data
(Lin et al., 2009)	EEG-based emotion recognition in music listening: A comparison of schemes for multiclass support vector machine
(Müller et al., 2008)	Machine learning for real-time single-trial EEG-analysis: From brain-computer interfacing to mental state monitoring
(Bos, 2006)	EEG-based Emotion Recognition

the lowest number of electrodes and the lowest price of \$199, offering a neurofeedback technology that was mainly designed for meditation and sleep, albeit this does not bound its research capabilities. The number of affordable EEG devices is increasing in the current market due to the Brain-Computer Interface (BCI) technology's noticeable progress, which led to a giant leap in the

consumer-grade segment. Consumer-grade products offer a lower price, accessibility, and simple design and allow researchers to readily study human brain activity without prior knowledge of electronics and engineering. The aforementioned EEG devices by three manufacturers, Emotiv, OpenBCI, and NeuroSky, are described in detail in the next section.

2.1. Emotiv

Emotiv is one of the biggest manufacturers of consumer-grade EEG devices. There are four Emotiv products available in the market: EPOC X, EPOC+, EPOC FLEX, and INSIGHT (“[Emotiv | Brain Data Measuring Hardware and Software Solutions](#),” n.d.). EPOC X is the latest product, released in 2020 in honor of the 10th-anniversary, that offers the features of the famous precursor model EPOC+. EPOC X model comes with an improved design and enables neuroscience research on a broader scope. There are 14 electrodes for tracking the cerebral cortex's electrical activity (+2 reference sensors located at the left and right mastoid process). Key features include wireless connectivity via Bluetooth, improved signal quality, comfortable design with rotating headband, motion sensors that can detect head movements, and robust saline-based electrodes (“[Emotiv EPOC X - 14 Channel Wireless EEG Headset | Emotiv](#),” n.d.).

EPOC+ is designed mainly for academic purposes. It has distributed sensors all across the scalp, spanning as much area as possible, and offering a simple assembly/disassembly (“[Emotiv EPOC+ 14-Channel Wireless EEG Headset | Emotiv](#),” n.d.). The model is an upgraded version of Emotiv EPOC that is no longer available in the market. In addition to the regular EEG electrodes, EPOC+ offers another nine inbuilt motions electrodes, such as gyroscope, accelerometer, magnetometer, each type having three sensors. The gyroscope and accelerometer collect information about accelerations in all three directions and rotations around each axis, while the magnetometer monitors the actual position and orientation in space. In this way, minor movements over continuous time can be offset by extra magnetic field data, enabling users to track the actual changes in the position more clearly, even during very complex motions. INSIGHT model might be a good option for those who seek a more budget-friendly device (“[Insight Brainwear® 5 Channel Wireless EEG Headset | EMOTIV](#),” n.d.). It is designed with semi-dry polymer sensors and has five primary sensors and two reference sensors, and effectively detects brain data. The INSIGHT model is more intended for everyday use rather than for research applications. Being the most expensive model, EPOC FLEX differs from other models in that it is a research-grade EEG device. It is equipped with a higher number of sensors, including 34 Saline-soaked felt pads and a control box that can transmit data wirelessly, switching a typical gel electrode cap system into a mobile recording device (“[EPOC Flex - 32-Channel Wireless EEG Device | Emotiv](#),” n.d.).

Early EPOC devices were released with a software called TestBench for recording raw EEG data. A software development kit (SDK) and PureEEG software were available for building new programs using raw EEG data. Effective December 31, 2019, the support for TestBench, SDK, and PureEEG has been phased out, albeit more limited open-source SDK is accessible online at the manufacturer's Github page (“[Emotiv](#),” n.d.). For TestBench and PureEEG users, Emotiv offers an EmotivPRO solution for data collection and event marking. EmotivPRO provides flexible cloud storage and options for quick data analysis without data exporting. For earlier SDK version users, Emotiv encourages switching to the Cortex SDK. Cortex SDK is the latest API that combines EEG recording with third-party software, empowering researchers and practitioners to develop BCI software or create customized games.

Table 2

A summary of six affordable EEG devices.

Device	Release Year	Price (USD)	Electrodes	Sampling Rate (Hz)
Emotiv EPOC FLEX	2018	1699+	up to 32 (+2)*	128
Emotiv EPOC X	2020	799	14 (+2)*	128/256
Emotiv EPOC+	2013	699	14 (+2)*	128/256
Emotiv INSIGHT	2015	299	5 (+2)*	128
OpenBCI	2014	750/1080	8/16	250/125
NeuroSky MindWave	2018	199	1	512

* (+2) indicates the reference sensors located at the left and right mastoid process.

2.2. OpenBCI

OpenBCI products were first released in 2014, as open-source hardware and firmware bundled with an open-source Graphical User Interface (GUI) (“OpenBCI - Open Source Biosensing Tools (EEG, EMG, EKG, and more),” n.d.). Compared to other affordable EEG devices, OpenBCI products only operate as an amplifier. Sampling rates vary depending on the number of channels, Cyton Biosensing Board having 250 Hz for eight channels, Cyton+ Daisy Biosensing Board with 125 Hz for 16 channels data. Apart from the brain data, the OpenBCI boards collect the heart and muscles’ electrical activity. There are various headset options for purchasing, including an option with a Computer-Aided Drafting (CAD) of the headset, where the user can 3D print the headset for particular study objectives. OpenBCI also provides a program called Processing, which can be used to write an open-source application.

OpenBCI products are gaining popularity for experimental studies. Several works demonstrated the reliability and potential of EEG signals collected by the OpenBCI device. For instance, Frey (2016) showed that EEG data obtained from OpenBCI is adequate for real-time applications with fewer electrodes than research-grade EEG equipment. Another study by Notaro and Diamond (2018) evaluated online learners’ engagement. In the same year, Rahma and Nurhadi (2018) used the OpenBCI EEG device to measure learners’ concentration while reading a task. Shen et al. (2017) examined drivers’ drowsiness, whereas Silva et al. (2017) examined EEG data of chess players. Moreover, new JAVA-based EEG software that is developed. It can detect, display, and analyze EEG data for various subjects (Stylianios et al., 2019). Another study presented by Nguyen et al. (2018) used Light-weight In-ear BioSensing (LIBS) system that can unceasingly collect the electrical signals from the human brain, eyes, and muscles. Further findings are related to EEG-based emotion recognition, having different numbers of participants involved (Al-Galal et al., 2016; Javaid et al., 2015; Katsigiannis and Ramzan, 2018; Lakhan et al., 2019; Liu et al., 2010). Given such flexibility for data collection, the device can help researchers understand how human emotions are aroused and controlled.

2.3. NeuroSky

NeuroSky Inc. is a producer of Brain-Computer Interface technologies for consumer-grade applications founded in 2004 (“EEG - ECG - Biosensors,” n.d.). NeuroSky can be called the pioneers of consumer-grade EEG products. Their first product launched in 2007, which is a Neurosky MindSet. The splash of NeuroSky MindSet in the market entailed the release of further upgraded consumer-grade products such as MindWave, MindWave Mobile, MindWave Mobile plus, and MindWave Mobile 2. However, only MindWave Mobile 2 is currently available globally, while earlier models’ production stopped throughout the years.

The MindWave Mobile 2 model is a popular low-cost EEG device. Developers state that it can detect cognitive function data, such as attention and meditation states, with only one EEG elec-

trode. This electrode is located on the forehead, which records the electrical data from the human brain’s frontal lobe, and the ear clip is fixed on the ear to create grounding to process the electrical noise. Although NeuroSky devices are known for their affordable price, simplicity in use, and quick data processing, they have low sensitivity and performance quality due to only one electrode. These devices also hardly cancel noise compared to other commercial EEG devices. Notably, few studies have presented the use of custom-built devices by NeuroSky. Wei and Ma (2017) used a customized EEG device with a single dry electrode to evaluate Taiwanese kids’ and adults’ reading and focus levels. Lim et al. (2012) employed a custom-built device with two dry electrodes replacing a single default electrode. The study evaluated a training program for kids with ADHD and claimed that an extra electrode significantly improved device performance. Thus, researchers still incline to use OpenBCI or Emotiv over NeuroSky products.

3. Emotion and EEG characteristics

The expanding interest in EEG-based emotion recognition revealed a variety of strategies for EEG data collection and analysis. These strategies aim to reveal how emotions can be induced and detected and further applied to develop brain-machine systems. The reliability of currently available low-cost EEG devices has been evaluated based on accuracy, convenience, and data acquisition quality (Debener et al., 2012; Frey, 2016; Kanoga et al., 2019; Lakhan et al., 2019; Maskeliunas et al., 2016). This study has conducted a literature review to explore low-cost consumer-grade EEG devices’ current status and their performance in emotion recognition studies. The literature search was mainly conducted on popular citation databases such as Google Scholar, Springer Link Online Libraries, Multidisciplinary Digital Publishing Institute (MDPI) and Science Direct (Elsevier). Basic search settings were used to identify topics that fall into our research interest. The search terms and keywords were: affect or emotion; emotion recognition or classification; EEG or Electroencephalography; affordable or consumer-grade; OpenBCI; Emotiv; MindWave; and NeuroSky. The initial search was refined, covering published studies in the last five years from 2015 to 2020, and resulted in 319 articles, as presented in Fig. 1. We then refined results based on article type, excluding book chapters, meeting abstracts, post-graduate dissertations and review articles. The search was restricted to publications in the English language only. Then the number reduced to 54. After detailed revision based on the type of EEG device used in the experiment setting, the number of participants, emotional stimuli, assessed emotions, and machine learning techniques, eight publications were selected and presented in Table 3. Assuming that these studies likely to deliver state-of-the-art research in terms of applicability and reliability of consumer-grade EEG devices. The table also presents pre-processing techniques, extracted features, and machine learning algorithms adopted for each study, together with the accuracy results. We also list the challenges of the consumer-grade device. Two research-grade EEG device studies by Zheng et al. (2014)

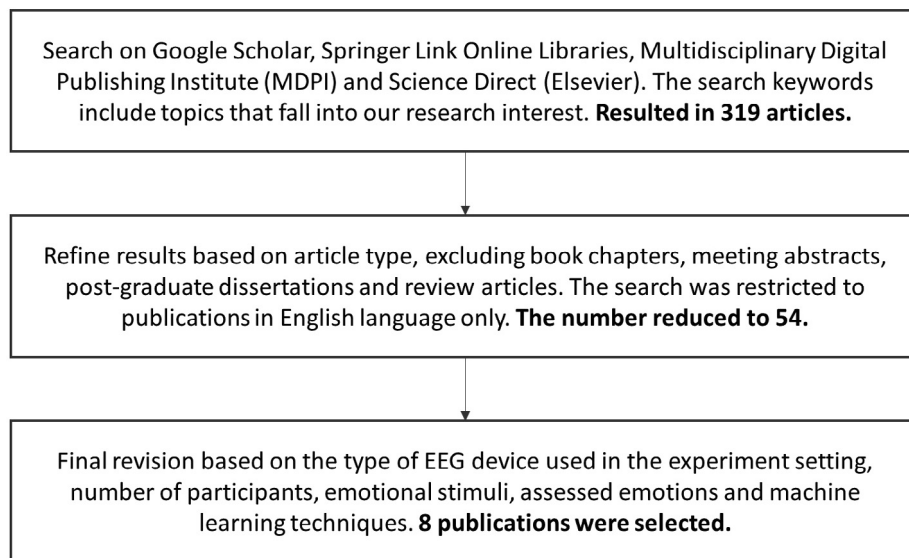


Fig. 1. The procedure of filtering and selecting articles.

and Zheng et al. (2019) are included in Table 3 to provide the differences in data quality and final results. These ten studies have shown that they assessed EEG data differently. They also used low-cost consumer-grade EEG devices to capture the data and used various machine learning classification models.

3.1. Emotion definition

There is no distinct definition of human emotion. It is a complex phenomenon, which is strongly related to a human's state of mind and feelings, including psychological, physical and social aspects (Ekman et al., 1987). Emotions can be described as changes in the human's brain state as a reflex towards a particular event. The affective state is a more common term used by researchers for emotional states (Picard, 2000). The study of human emotions has a long history and presently covers various areas such as psychology, computer science, neuroscience, medicine, sociology and philosophy (Ekman and Davidson, 1994). William James (1884), an American philosopher and originator of psychology, claimed that emotions are the feelings from the human body's changes. He suggested that any emotion results from a particular physical change inside the human body, such as muscle movement, facial expression and visceral motor activity. Similar ideas were presented by Lange (1885); thus, further investigations defined their concept as the James-Lange theory (Cannon, 1927). This theory gives a solid foundation for future research of human affective states based on physiological signals (Ekman, 2005).

3.2. Models of emotion

The generation of emotions activates different brain areas, and it is essential to know how to classify them. Although various researchers propose different emotion classification models, there is no consensus on the general model. The existing models fall under two categories, using a discrete definition for each emotion, or classifying emotions based on dimensional scales (Scherer, 2005).

Discrete emotional models classify emotions based on a small group of universal discrete emotions. They are called universal as they are common in different cultures, and some are applicable for primates and other mammals. Hence they are naturally cultivated (Panksepp, 1998). For decades, discrete emotion models included six emotional states: happy, sad, frightened, angry, sur-

prised, and disgusted (Ekman, 1992). The discrete emotional models are frequently applied in emotion recognition systems where the computer predicts several emotional states. However, these models are not capable of fully reflecting the complexity of emotions. Next, a dimensional emotion model is created to address emotions' complexity using different affective states' dimensions. Emotions are classified in this model as a set of scales. It is where each scale is a dimension of a multi-dimensional system. A dimension can be continuous or discrete, having a minimum and a maximum value. Emotions can then be interpreted and described based on each scale's values or points in a multi-dimensional expanse (Scherer, 2005). Such a way allows researchers to focus on emotion recognition tasks without concern about what category of emotion to follow. One of the commonly used dimensional models is Russel's circumplex model. It uses two dimensions to classify emotions. One is the valence scale, and the other is the arousal scale. Valence scale ables to capture the feeling of pleasant to unpleasant. On the other hand, the arousal scale captures the feeling of active to inactive (Russell, 1980), as shown in Fig. 2.

3.3. Electroencephalography

EEG is a process of monitoring and recording information about the electrical activity of the human brain. EEG signals are formed by spontaneous and rhythmic impulses of the brain neurons. In neuroscience and psychology, it is suggested that EEG signals can describe affective brain states and human behavior. (Ackermann et al., 2016). EEG is frequently applied in emotion recognition research (Bozhkov et al., 2015; Liu and Sourina, 2014; Wang et al., 2019). Subtle fluctuations in human affective states can be captured by EEG signals (Dietrich and Kanso, 2010). However, EEG signals are weak and difficult to record. They can be easily intervened by other physiological signals, arising from electrocardiography (ECG), electrooculography (EOG) and electromyography (EMG) (Subha et al., 2010). Thus, the structure of EEG signals is non-linear and chaotic (Garrett et al., 2003). Raw EEG signals typically undergo denoising and pre-processing.

3.4. EEG-based emotion recognition process

Past studies (Bairavi and Sundhara Kumar, 2018; Blanco et al., 2019; Lakhan et al., 2019; Müller and Fritz, 2015; Zheng

Table 3

Summary of recent emotion recognition experimental studies from EEG devices.

Reference	Device	Participants (Female/ Male)	Stimuli	Assessed Emotions	Number of Electrodes	Pre- processing	Features Extraction	Classification Algorithm	Accuracy Results
Lakhan et al. (2019)	OpenBCI	43 (21/22)	audio-visual (film trailers)	valence (high/ low) arousal (high/ low)	8	CAR, Notch filter, ICA	PSD	K-Means Clustering	67.44% for arousal 66.67% for valence
Javaid et al. (2015)	OpenBCI	8 (1/7)	audio-visual	valence (high/ low) arousal (high/ low)	3 & 8	Notch filter, bandpass filter	Higuchi's FD	SVM-RBF	3 electrodes case: 59.1% for arousal 68.39% for valence 8 electrodes case: 87.62% for arousal 83.28% for valence
Müller and Fritz (2015)	NeuroSky MindWave	17 (1/16)	subject-induced (performing change task)	valence (high/ low) arousal (high/ low)	1	bandpass filter	PSD	Decision Tree	71.36% for positive and negative emotions
Bairavi and Sundhara Kumar (2018)	NeuroSky MindWave	18 children	audio (music clips)	valence (high/ low) arousal (high/ low)	1	bandpass filter	PSD	Random Forest SVM-Sigmoid Naïve Bayes KNN	82% 71% 67% 56%
Zheng et al. (2016)	NeuroSky MindWave	20	subject-induced (riding a stationary bike)	anxiety (high/moderate/ low)	1	bandpass filter	Partial autocorrelation, mobility, complexity, mean of wavelet coefficients, power of wavelet coefficients	SVM-RBF	62.5% 50%
Blanco et al. (2019)	Emotiv EPOC	18 (all male)	visual (Stroop color– word interference test)	stressed (high/ low)	14 (+2)	bandpass filter, FIR filters	RMS	Logistic Regression QDA KNN	70.72% 68.63% 66.09%
Katsigiannis and Ramzan (2018)	Emotiv EPOC	23 (9/14)	audio-visual (film clips)	valence (high/ low) arousal (high/ low) dominance (high/low) valence (high/ low) arousal (high/ low)	14 (+2)	FIR filters, ASR, CAR	PSD	SVM-RBF	62.49% for Valence 62.17% for Arousal 61.84% for Dominance 65.09% for negative emotions 86.43% for positive emotions
Liu et al. (2018)	Emotiv EPOC	30 (all male)	audio-visual (film clips)	valence (high/ low) arousal (high/ low)	14 (+2)	bandpass filter, ICA	PSD	SVM-RBF	70.43% 84.08% 78.21% 83.26% 91.07%
Zheng et al. (2019)	ESI NeuroScan	15 (8/7)	audio-visual (film clips)	valence (high/ low) arousal (high/ low)	62	bandpass filter	PSD, differential entropy, differential asymmetry, rational asymmetry, differential caudality	KNN Linear Regression SVM-RBF SVM-Linear GELM	70.43% 84.08% 78.21% 83.26% 91.07%
Zheng et al. (2014)	ESI NeuroScan	6 (3/3)	audio-visual (film clips)	valence (high/ low) arousal (high/ low)	62	bandpass filter	Differential entropy	DBN-HMM DBN GELM SVM-Linear KNN	87.62% 86.91% 85.67% 84.08% 69.66%

Note: Support Vector Machine (SVM), Radial Basis Function (RBF), K-Nearest Neighbors (KNN), Quadratic Discriminant Analysis (QDA), Power Spectral Density (PSD), Fractal Dimension (FD), Root Mean Squared (RMS), Deep Belief Network (DBN), Hidden Markov Model (HMM), Graph regularized Extreme Learning Machine (GELM), Independent Component Analysis (ICA), Finite Impulse Response (FIR), Artifact Subspace Reconstruction (ASR), Common Average Reference (CAR).

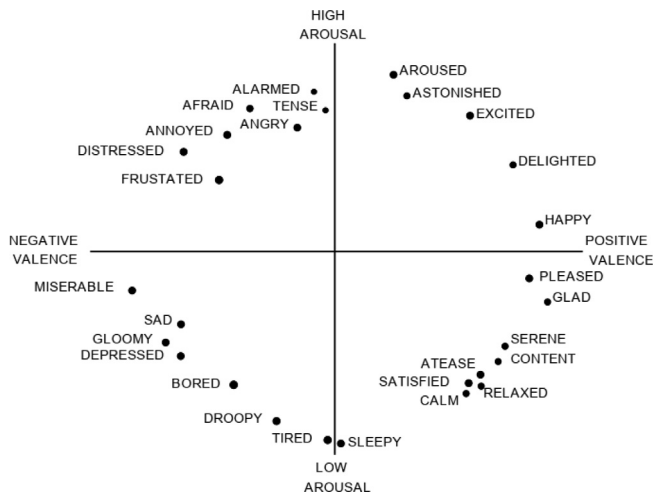


Fig. 2. Russell Valence-Arousal Model (Russell, 1980).

et al., 2014, 2016; Javaid et al., 2015; Katsigiannis and Ramzan, 2018; Liu et al., 2018; Zheng et al., 2019) have shown six main EEG-based emotion recognition procedure. There are participants and stimuli selection, signal acquisition, pre-processing, feature extraction and classification. This process is depicted in Fig. 3.

The first two steps of the EEG-based emotion recognition process are recruiting participants and selecting the powerful stimuli required for emotion elicitation. We did not find any evidence on generic instruction to determine the highly effective stimuli from the reviews. Many studies have shown that the participant selection process is closely related to the experiment's stimuli. The selection process mainly relies on an individual's background, gender, age, character, and other personal factors.

In the signal acquisition step, the participants are required to wear an EEG device. They are exposed to the experimental stimuli (audio/video), and the voltage fluctuations spotted in the brain of the participants are collected. EEG does not record the activity of neurons separately but instead records the signals from the population of neurons when they are active all at once. It mainly collects data from small areas of the brain surrounding each electrode. EEG presents an image of electrical activity in the brain built of waves of varying frequency, amplitude and shape. Next, the data will go through a pre-processing stage where the external artifacts and interferences are filtered from the obtained EEG data, including noise reduction and spatial and temporal filtering. The clean data is analyzed and followed by a feature extraction process. Feature extraction is one of the most critical steps in an EEG-based emotion recognition system. There is no distinct clarity of which feature represents which emotion (Jenke et al., 2014). The features highly related to the emotional states such as power spectral density, fractal dimension values and differential entropy are extracted from the pre-processed EEG data. Lastly, a machine learning classifier is used to train and interpret the EEG data.

3.5. Participants

The number of participants may vary depending on the study type and study area. In the reviewed studies listed in Table 3, the

number of participants involved ranges from 6 to 43. The majority of the studies had a reasonably large number of participants to enhance the experimental solidity. Two of the experimental studies (Javaid et al., 2015; Zheng et al., 2014) had less than 10 participants, and only four (Katsigiannis and Ramzan, 2018; Lakhan et al., 2019; Liu et al., 2018; Zheng et al., 2016) had at least 20 participants. The other four studies involved several participants ranging from 15 to 18 (Bairavi and Sundhara Kumar, 2018; Blanco et al., 2019; Müller and Fritz, 2015; Zheng et al., 2019).

From Table 3, only four studies considered gender factors when selecting participants (Lakhan et al., 2019; Zheng et al., 2014; Katsigiannis and Ramzan, 2018; Zheng et al., 2019). According to Brody and Hall (2008), the gender differences in emotion expression happen due to biologically-based temperamental predispositions and socialization to adopt gender-related rules to express emotions. It is claimed that there are gender-related display rules in various cultures for females to be more emotionally expressive than males (Kring and Gordon, 1998). In other words, females are assumed to express a higher level of emotional states than males, specifically happiness and imputing negative emotions, such as guilt, anxiety, fear, sadness, and shame (Brody and Hall, 2008). It was also claimed that emotion recognition accuracy is better for females than males in the middle and older ages (Demenescu et al., 2014). Based on these findings, it can be deduced that gender and age factors are significant when dealing with emotion recognition systems. They should be considered a recruitment attribute in EEG-based emotion recognition studies to examine how emotions are expressed for different gender and ages.

3.6. Stimuli

According to Yu Chen and Mehmood (2019), emotions can be induced by a subject-induced or event-induced approach. The subject-induced approach requires the participant to recall the emotion from the early data collection experiment and then remember it. In contrast, the event-induced approach uses various modalities to elicit the emotion of the subject. Audio and video files are commonly used. Although some studies use mental imagery to induce emotions (Blanco et al., 2019; Zheng et al., 2016), most of the studies rely on audio-visual stimuli, such as pictures, music clips, and videos to evoke specific emotions (Bairavi and Sundhara Kumar, 2018; Katsigiannis and Ramzan, 2018; Lakhan et al., 2019). Audio-visual stimuli are mainly selected based on dimensional emotional models (Chanel et al., 2009). A study by Petrantonakis and Hadjileontiadis (2010) has used visual stimuli and facial expression recordings to detect basic emotions of sadness, happiness, disgust, fear, surprise and anger. A study by Lin et al. (2010) collected EEG data while participants were exposed to audio stimuli. The obtained brain signals were further categorized into pleasure, joy, anger and sadness. Sometimes researchers use a combination of audio and video stimuli, like in a study by Nie et al. (2011), where 12 film clips were thoroughly selected to induce positive and negative emotions. Compared to other emotion stimulation methods, the film clips offer solid emotional context, dynamic and biologically suitable stimuli within an adequately short duration, and standardization simplicity. In a study by Coan et al. (2006), participants were asked to express their emotions by mimics, assuming that human emotions can be

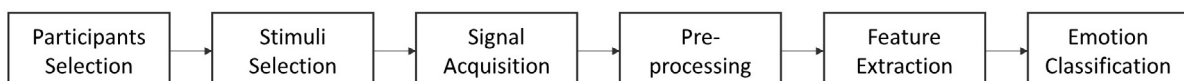


Fig. 3. Process of emotion detection from EEG signals.

revealed through facial expression. However, this approach is unreliable, as a human can quickly adopt and fake their inner emotions. [Kothe et al. \(2013\)](#) asked participants to recall memories that can induce certain emotional states, and EEG data were recorded simultaneously. This approach is entirely in control of participants, and some studies refer to it as the primary way of investigation ([Damasio et al., 2000](#)). Nevertheless, it can be troublesome for a participant to recall suitable memory fragments with corresponding emotion, leading to unreliable results.

Some studies use self-selected emotional stimuli. For instance, [Lakhan et al. \(2019\)](#) used 120 emotionally-charged trailer clips selected from the Internet Movie Database (“IMDb: Ratings, Reviews, and Where to Watch the Best Movies & TV Shows,” n.d.) to evaluate the applicability of OpenBCI, a consumer-grade EEG device for emotion recognition. In the same year, [Blanco et al. \(2019\)](#) used a Stroop color-word interference test to induce mild stress on their subject. Visual stimuli (color-words) were produced using Microsoft Paint to stimulate stress responses. [Zheng et al. \(2016\)](#) adopted similar stimuli. They instructed participants to ride a stationary bike and imagine that rivals were overtaking them to induce anxiety emotions. The results of these three studies indicate that these stimuli are capable of inducing targeted emotions.

There are few publicly available stimuli databases, particularly the International Affective Picture System (IAPS) and International Affective Digitized Sounds (IADS). Both comprise a comprehensive set of images and sounds with a labeled rate for valence, arousal and dominance level evaluated by 100 participants ([Bradley and Lang, 2007](#)). Notably, auditory stimuli are less popular even though it was proven to be equally effective as visual stimuli ([Zhou et al., 2014](#)). The advantage of using publicly available stimuli datasets is that researchers do not need to evaluate and sort out effective stimuli for their experiments. All the stimuli are already validated and ready to be used for EEG data collection.

3.7. Placement of electrodes

An important aspect of emotion recognition study is electrode placement. There is a standardized technique to place the electrodes across the scalp, and this technique is performed according to the 10–20 international system ([Herwig et al., 2003](#)). The numbers “10” and “20” signify the actual distances between the adjacent electrodes, which are 10% or 20% of the entire distance from front to back or left to right of the brain, and each electrode has a label that helps the researcher to identify it ([Nicolas-Alonso](#)

and [Gomez-Gil, 2012](#)). Additional electrodes can be located on any of the available empty spots. Another variation of this system is higher resolution, called the 10–10 international system, with only 10% of the adjacent electrodes’ distance. [Fig. 4](#) demonstrates the electrode’s placement based on the 10–20 international system, while [Fig. 5](#) shows the arrangement of electrodes based on the 10–10 international system. Each electrode has a unique label composed of one or two letters followed by a number. Those letters correlate with the brain area where the electrode is placed, for instance, the letter T for temporal, O for occipital, F for frontal, P for parietal, and C for central. Even numbers are given for electrodes on the right hemisphere, while odd numbers are given for the electrodes located on the left hemisphere.

All electrodes located on the right hemisphere have the exact opposite electrodes on the left side, with a label number increased by one. These electrode pairs are referred to as symmetrical. In the reviewed studies, the number of electrodes ranges from 3 to 62, and they were placed according to the 10–20 system. Three studies ([Bairavi and Sundhara Kumar, 2018](#); [Müller and Fritz, 2015](#); [Zheng et al., 2016](#)) used NeuroSky MindWave Mobile. As mentioned earlier in section 2.3, the main drawback associated with the NeuroSky devices is only one electrode located at the forehead. It leads to low sensitivity as well as inflexibility, which affects product usability. For EEG-based emotion recognition, the perfect combination would be fewer electrodes with high data quality and comfortability.

3.8. Pre-processing

The foremost step for EEG-based prediction of user emotions is to remove unwanted noises and artifacts, after which EEG signals are categorized based on rhythmic activity. The EEG signals have a low signal-to-noise ratio (SNR); thus, they might be easily disturbed by noise. Artifacts may occur from motion, such as head or an eye movement or even heartbeat, which sometimes has a higher wave range than brain waves. The pre-processing stage deals with removing unwanted artifacts, such as EOG, ECG, EMG interferences, and electromagnetic and frequency types of interferences. Various methods have been presented to eliminate EOG and EMG-related noise ([Brunia et al., 1989](#); [Gratton et al., 1983](#); [Wallstrom et al., 2004](#)). The electromagnetic and frequency interferences mainly emerge at high frequencies, so it is an apt approach to use low-pass or band-pass filtering to cut off unwanted frequency bands and only keep EEG-related frequency. The pre-processing step’s primary function is to eliminate

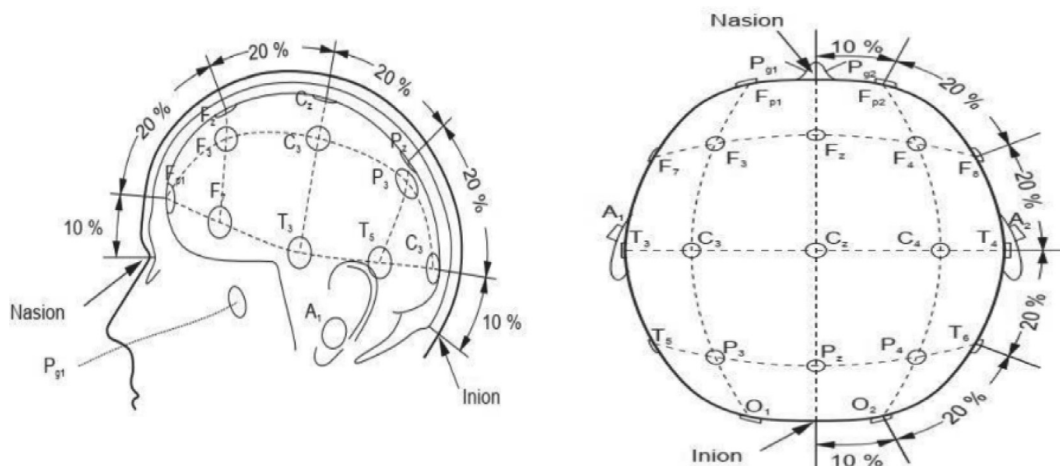


Fig. 4. The 10–20 electrodes placement system ([Nicolas-Alonso and Gomez-Gil, 2012](#)).

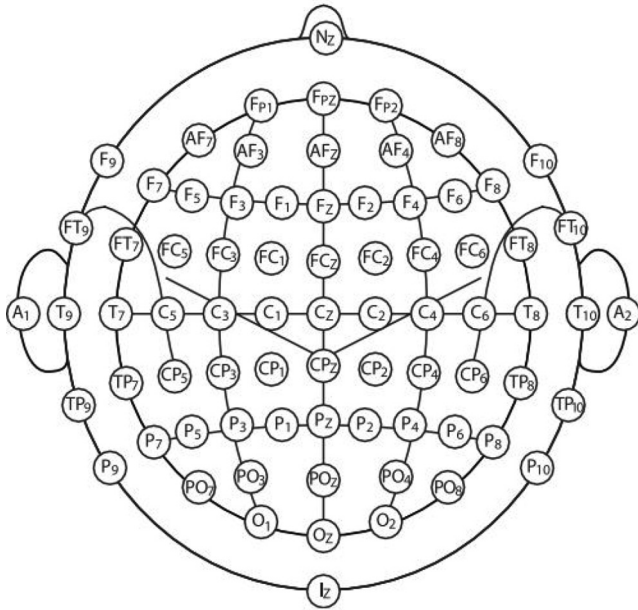


Fig. 5. The 10–10 electrodes placement system (Acharya et al., 2016).

information that is not relevant to emotional patterns and affect the final results. Pre-processing of EEG signals involves several steps, such as downsampling, line noise removal, bandpass filtering, and removal of weak EEG channels. Additionally, it contributes to the efficient extraction of distinct features. Most of the studies reviewed in this paper conducted simple filtering to eliminate power-line noise and drifts by applying a notch or bandpass filter (Bairavi and Sundhara Kumar, 2018; Müller and Fritz, 2015; Zheng et al., 2014, 2016; Javaid et al., 2015; Zheng et al., 2019). Researchers have reported that if the participants whose signals were overly noisy, then a decision has to be made whether to remove or retain the sample. For instance, the removal of signal noise from eye blinking and muscle movement using ICA and FIR filters (Blanco et al., 2019; Katsigiannis and Ramzan, 2018; Lakhani et al., 2019; Liu et al., 2018).

The filtering process cannot eliminate all the artifacts from the signals, and therefore, additional processing is sometimes required. The Artifact Subspace Reconstruction (ASR) and Common Average Reference (CAR) techniques, introduced by (Kothe, 2016) and Cohen (2014), respectively, were additionally applied in a study by Katsigiannis and Ramzan (2018). ASR is an automatic, component-based technique that can efficiently eliminate large-scale or transient artifacts contaminating EEG signals. On the other hand, the CAR technique estimates the mean value through all electrodes and deducts it from each electrode sample. CAR is mainly used in EEG when required to detect small signal sources in highly noisy data (Ludwig et al., 2009). It is necessary to differentiate between the various sources of signals that can disturb the data. However, some of these artifacts might contain pertinent emotional information and help get higher performance accuracy.

3.9. Feature extraction

Feature Extraction deals with extracting the specific feature from the EEG signal, which is valuable for the study objectives. It is an essential part of EEG-based emotion recognition. It aims to compute the reliable feature conforming to the emotions of interest to assure satisfactory classification results. The perfect feature would facilitate an adequate classification of affective states and define the EEG-based emotion recognition system (Kim et al., 2013). Generally, EEG features extraction methods are performed based on the frequency domain (Valdes et al., 1992), time-domain (Hjorth, 1970), and time-frequency domain (Roach and Mathalon, 2008).

Frequency-domain features are obtained by converting the original EEG data from a time domain to a frequency domain and then compute the valuable frequency information as EEG features. Generally, five frequencies are used by researchers, such as delta, theta, alpha, beta and gamma bands, as shown in Fig. 6. Frequency bands can be used to describe various functions of the human brain, such as emotional state, cognitive load, problem-solving, mediation, and many more (Klimesch, 1999). Delta waves, ranging from 0 to 4 Hz, are typically slow and are associated with deep sleep. It is mainly observed in infants. However, it is not a standard component of EEG data of awake teenagers and adults. Theta waves, ranging from 4 to 8 Hz, are relatively slow and observed

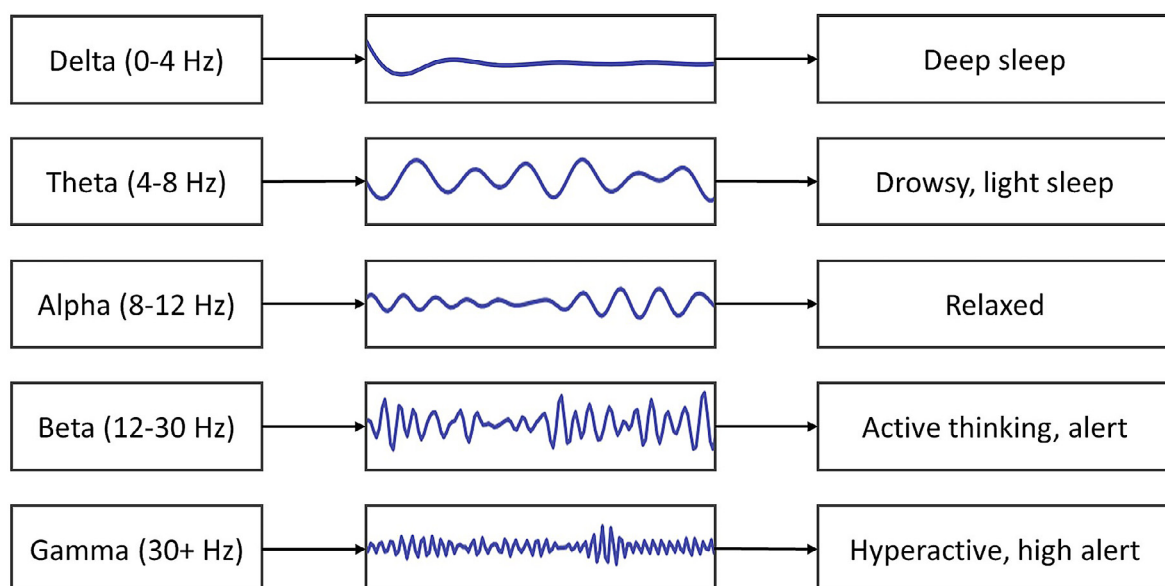


Fig. 6. EEG frequency bands and associated mental states.

in adults when they are drowsy, relaxed, or meditating. Alpha waves, ranging from 8 to 13 Hz, are generally associated with visual processing, mental load, and memory activities. Beta waves, ranging from 13 to 30 Hz, are linked to the brain's excited or alert state, including active thinking and close attention. High-level beta waves might be reflected when people are frightened or panicked. Above 30 Hz, Gamma waves are high-frequency waves. They are associated with high-level brain activity in the brainstem, such as information perception, information analysis and visual/auditory stimuli synthesis. There are several methods available to convert EEG data from time-domain to frequency-domain, such as Discrete Fourier Transform (DFT) and Fast Fourier Transform (FFT) (Nie et al., 2011; Wang et al., 2011). The power spectrum, power density spectrum and frequency band power are widely used frequency-domain features (Jenke et al., 2014). Computation of frequency-domain characteristics of EEG signal mainly originates from the Power Spectral Density (PSD) evaluation. The PSD value can be obtained based on the FFT of a particular time-series EEG signal, as the ratio of the mean average squared amplitude frequency to time feature (Kim et al., 2013).

The time-domain features are computed from EEG signals to reveal the time-domain data or time-related statistics as a feature (Hjorth, 1970). EEG is the typical continuous time-series data, and it contains highly intuitive information to describe human emotions. Computation of time-domain features is a relatively straightforward process in EEG-based studies. It uses the original pre-processed time-series signals, meaning that no transformation has been done compared to other types of feature extraction methods. Thus, time-domain EEG studies are considered fundamental and high-performing (Koenig et al., 2005). An excellent example of time-domain features is statistical features by considering mean, variance, standard deviation, skewness, peak, and many more. (Jenke, Peer and Buss, 2014). Another classic time-domain feature is Higher-Order Crossing (HOC), which applies the zero-crossing counting filter to the time series EEG signal (McKenzie and Kedem, 1996). Fractal Dimension (FD) is another renowned mathematical method used to extract the time-domain features (Pradhan and Narayana Dutt, 1993). Various studies verified FD features' effect, considering the non-linear and chaotic EEG signal structure (Kulish et al., 2006; Lutzenberger et al., 1992).

When the signal is unstable, time–frequency features can provide valuable characteristics based on the dynamical fluctuations (Jenke et al., 2014). Since EEG signals are non-stationary, assuming that time-domain or frequency-domain features are insufficient, various EEG-based time–frequency domain characteristics have been analyzed to extract information from both time and frequency domains (Alazrai et al., 2018; Chanel et al., 2009). Furthermore, the core idea of time–frequency domain features is the dynamic fluctuations in time-series, which conforms to the main objective of emotion classification study. Since human emotions' physiological aspects are based on time, after a particular emotional state being elicited, it will exist for a certain time and then return to a normal calm state (Ekman et al., 1987). From this point of view, the time–frequency domain features can bring more valuable information. Some of the widely used techniques for time–frequency domain feature extraction are Discrete Wavelet Transform (DWT) (Bhattacharyya et al., 2018; Kamal Al-Qazzaz et al., 2017) and Short-Time Fourier Transform (STFT) (Rayatdoost and Soleymani, 2018; Saifutdinova et al., 2016).

As presented in Table 3, there are various feature extraction techniques commonly used by researchers. One of the widely used features is obtained from the power spectrum of the collected EEG data (Bairavi and Sundhara Kumar, 2018; Katsigiannis and Ramzan, 2018; Lakhan et al., 2019; Liu et al., 2018). The spectral analysis focuses on the power spectrum at various frequency bands employing the discrete Fourier transform. In contrast, Javaid et al.

(2015) has used Higuchi's Fractal Dimension instead, which is suitable for analyzing the nonlinear systems and could be used in real-time EEG signal processing. Another frequently used class of features is differential entropy Zheng et al. (2014) and Zheng et al. (2019). This feature expands Shannon entropy's idea and can measure the complexity of a continuous random variable. It was revealed that differential entropy for a certain length of EEG wave is similar to the logarithm energy spectrum in a specific frequency band (Duan et al., 2013). Notably, a single feature extraction method is not optimal for all of the applications. Besides, the number of signals collected might be insufficient for high-quality feature extraction. Different approaches might bring in more features to reveal supplementary data about the human brain, like in a study by Zheng et al. (2016) and Zheng et al. (2019), where they have used more than one feature.

Thus far, no studies highlight an emotion-related EEG feature that can analyze all features. Indicators from past research have shown that the power features method can efficiently identify various bands (Jatupaiboon et al., 2013; Pan et al., 2020; Thammasan et al., 2015). Lakhan et al. (2019), Liu et al. (2018), Müller and Fritz (2015) are the researchers who affirm that PSD features extracted are suitable to recognize EEG data and differentiate the emotional states. Once an optimal feature that can effectively differentiate the emotional states is selected, a classification algorithm can then be trained to create an emotion detection system.

3.10. Emotion classification

The emotion classification step deals with detecting different emotional states based on the information provided by the extracted features (Kim et al., 2013). Machine learning algorithms are applied for this step, as they are highly efficient for multi-dimensional EEG features. There are two types of algorithms introduced in emotion classification, classifier and regressor. A classifier identifies a given observation class from a set of classes, while a regressor algorithm can assess patterns output as a dependent variable. In other words, if emotions are denoted as separated classes or the emotional levels are divided into a set of scales, then a classification algorithm is more suitable. Still, if emotions are denoted as multi-dimensional levels, and each level is referred to as a continuous variable, then a regression algorithm would be preferred.

EEG classification refers to getting the observations to identify specific emotional states from the given EEG features (Kim et al., 2013). The EEG features are fed into a machine learning classifier for training the principles of the corresponding affective state under various algorithms. An EEG-based emotion classifier's two main goals are identifying the patterns of various emotions through EEG features; to categorize novel untrained feature samples into appropriate EEG patterns. There are two types of machine learning classification methods, namely supervised and unsupervised. Supervised learning requires labeled sample classes and constant revision of model specifications with guidance from class information. In other words, it is a model created by using labeled training data that contains input data and a targeted output. The label for each class is the crucial factor for supervised learning. There are two common labeling strategies for EEG data. One way is when emotional states can be defined by the dimensional model, such as arousal-valence dimensions, originated from the psychology field (Ekman et al., 1987). Various affective states can be designated by combining different valence levels and arousal levels (Koelstra et al., 2012; Sourina and Liu, 2011). Another method is defining a specific emotional state of a subject based on the stimuli used throughout the data collection process. This method can be further supported by monitoring the subject's facial expressions, heart rate, and body gestures to identify the emotional state

(Zheng and Lu, 2017). Support Vector Machines (SVM), Decision Trees, Random Forest, Bayesian Network, Naive Bayes, K-Nearest Neighbors (KNN) and deep neural networks are examples of supervised learning models (Kim et al., 2013).

Unsupervised learning is a type of algorithm that draws inferences from the unlabeled input features. It will perform training on the sample data without any information about classes. In other words, it lets the model detect different patterns within features on its own. Hence, unsupervised learning will identify patterns from the unstructured input data by monitoring the relationships between the features and group alike features identified (Gath and Geva, 1989). Unsupervised learning algorithms are widely used in EEG-based emotion recognition research, as human affective states are complex, and there is no standard procedure to define emotions empirically. Generally, clustering, density estimation, and dimensionality reduction tasks are referred to as unsupervised learning. The unsupervised learning models are complicated than supervised learning, as it gets insufficient or no information about the data characteristics. Some of the renowned unsupervised learning techniques are K-means clustering and fuzzy clustering (Gath and Geva, 1989; Rai et al., 2015).

3.11. Summary of reviewed studies

In both studies conducted by Katsigiannis and Ramzan (2018) and Liu et al. (2018), the Emotiv EPOC device was used to collect data while subjects were exposed to emotionally-charged clips. Katsigiannis and Ramzan (2018) obtained EEG and ECG data together with a self-assessment survey on arousal, valence and dominance. The classification method was constructed based on SVM-RBF and tested on EEG features and ECG features separately. Besides, the fusion of these features was tested as well. EEG features alone were higher than ECG, having 62.49% for Valence, 62.17% for Arousal, and 61.84% for dominance. Fusion results showed nearly similar accuracy of 61.84%, 62.32%, and 61.84% for valence, arousal, and dominance, respectively. On the other hand, Liu et al. (2018) introduced a real-time emotion recognition model that works on EEG data straight fed from the EEG device. This chain is further hooked with an EEGLAB toolbox (“EEGLAB,” n.d.), which is a built-in Matlab software (“MATLAB - MathWorks - MATLAB & Simulink,” n.d.) that further aids with signal processing, feature extraction, and classification steps. The proposed method consists of 3 layers of SVM-based classification methods. The first layer detects neutral and non-neutral emotions. The second layer distinguishes the previous layer's results into positive and negative emotions. The last third layer separates emotions like joy, tenderness, and amusement for positive groups, sadness, anger, disgust, and fear for negative groups. The accuracy among all layers ranges from 65% to 92%.

Lakhan et al. (2019) examined the OpenBCI device's feasibility in an emotion recognition study. They have recruited 43 individuals to watch pre-labeled emotional visual stimuli. As the subjects finished watching the visual stimuli, they were asked to rate each stimulus on a scale of 1–9 and evaluate the level of happiness, excitement and fear. The ratings are further used to label the data. The collected signals from OpenBCI and Empatica4 or E4 (device to collect peripheral physiological signals) were embedded into classification algorithms based on K-means clustering for training and testing. The results revealed that this approach could provide an accuracy of 70% for arousal and 67% for valence prediction. This work was the former of presenting validation of OpenBCI for emotion detection algorithms and able to deduce that OpenBCI equipment is sufficient to apply in emotion recognition studies. Javaid et al. (2015) conducted a comparative study by confronting the effect of using three and eight electrodes across the scalp. The OpenBCI device was used for EEG data collection. SVM-RBF algo-

rithm is used for classification. Results revealed that using eight electrodes outperforms three electrodes. The minimum performance precision of the SVM algorithm of the four emotions, namely pleasant, sad, happy and frustrated, has reached up to 75%. Specifically for arousal recognition, accuracy jumped from 59.17% to 87.62%, while valence results show a range of 68.39% to 83.28%.

A study by Zheng et al. (2016) collected EEG data using a NeuroSky MindWave Mobile device and photoplethysmogram (PPG) signals using PPG built-in glasses. The subjects were instructed to ride a stationary bike, start pedaling at a leisurely speed and then start increasing the speed as if they were competing while imagining their rivals speeding up to 80 km/h. Eliciting a feeling of anxiety by creating a competing environment was targeted in this experiment. At the beginning of each session, an individual's baseline anxiety score was evaluated with a Competitive State Anxiety Inventory-2 (CSAI-2) test. For classification, features extracted from EEG and PPG data were fed into algorithms built based on KNN and SVM-RBF. The results have shown the classification accuracy of only 50% and 63% for SVM-RBF and KNN, respectively. In a separate study, Blanco et al. (2019) show that low-cost Emotiv EPOC device present acceptable accuracy in detecting two types of cognitive stress states on a single-trial basis. They have experimented in a controlled setting. Using a logistic regression algorithm to analyze the data, they have obtained a 70.7% mean accuracy for all participants at the Fc5 electrode. Müller and Fritz (2015) investigated 17 volunteer developers' emotions while performing a software change task. A decision tree classifier was used to train and implement via WEKA software (“Weka 3 - Data Mining with Open Source Machine Learning Software in Java,” n.d.). The decision tree is a supervised classification method built on the segmentation of a complex problem into smaller groups. They were iterating this process periodically using a tree generation. Later, every leaf node is given a class label, nonterminal nodes, which enclose the root and other internal nodes, having feature testing conditions to segregate data that exhibit various attributes (Tan et al., 2006). As a result, this classifier was able to detect positive and negative emotions with an accuracy of 71.36%.

A study by Zheng et al. (2014) tried to differentiate between several pattern recognition algorithms to detect the best option using ESI NeuroScan equipment (“Compumedics Neuroscan – World Leader in Functional Neuro-imaging,” n.d.). The study involved 6 participants, and EEG data were classified using Deep Belief Network (DBN) with differential entropy features extracted from 62 channel EEG as input. The Hidden Markov Model (HMM) was fused with DBN to improve accuracy. Results obtained were compared to KNN, SVM-Linear and Graph regularized Extreme Learning Machine (GELM) algorithms. They were able to achieve 87.62% and 86.91% accuracy for DBN-HMM and DBN, respectively. Later, Zheng et al. (2019) conducted a similar study, using the same device, but this time having 15 participants and examining more features and classification algorithms. Best accuracy was achieved by linear regression having 84.08% and GELM classifier having 91.07% accuracy. Zheng et al. (2019) have also investigated how stable their emotion recognition model across participants and trials. It was shown that the model performance across participants and trials is worse than that on a single subject-based experiment. Ideally, the model is trained on the features from a group of participants or trials and predicts the new data from other unknown participants or trials. These types of models are referred to as subject-independent. Subject-independent models are technically troublesome due to individual differences across participants with the EEG signals' inherent variability (Olivetti et al., 2014). Emotions have some individual differences for different participants, which may bring changes in an underlying probability distribution. That is why the average accuracy of the subject-dependent or trial-

based classification models is much higher than that of models trained on a set of participants or trials and tested on other new participants.

Discussions

Based on the systematic review method, we identify eight studies that examined commercial EEG devices for emotion recognition between 2015 and 2020. Table 3 summarizes these studies together with two studies that used research-grade EEG devices. Adding this allows us to compare the accuracy rate among research-grade and consumer-grade devices. Table 3 lists the type of EEG devices used, the number of participants involved, stimuli used, emotions examined, pre-processing methods used, features extracted, type of machine learning classifiers used, and the final emotion classification accuracy of all ten studies.

The current study found that various emotional stimuli have been used and reported. The most popular technique among researchers is event-induced emotion stimulation, where the participants passively perceive emotional stimuli such as images, music, and videos to evoke certain emotions. The event-based elicitation technique's main advantage is that the stimuli can be significantly standardized, and emotions can be manipulated appropriately. Subject-induced elicitation is another popular method, where participants are instructed to perform a specific task that can naturally evoke different emotional states. For instance, participants are asked to recall specific memory segments from their own experiences associated with a corresponding emotion. This process is entirely under the control of participants, and some studies treat it as an essential method of research (Damasio et al., 2000). The lack of subject-induced technique is that participants might not find the required memory segment for a given emotion and measurement of emotion retention time might be challenging. Nevertheless, this technique can evoke more naturalistic emotions similar to the affective states occurring in real life.

Several studies used the subject-induced elicitation method, such as imagining competing with another person while riding a bike (Zheng et al., 2016), performing software change tasks (Müller and Fritz, 2015), and taking neuropsychological tests (Blanco et al., 2019). The accuracy results suggest that standardized stimuli, primarily audio-visual stimuli, are more effective than a mental change task or imagination stimuli. The fewer number of electrodes can facilitate the comfortability of the user, system usability, and time spent to set up an EEG device; for instance, only one electrode was used in three studies (Bairavi and Sundhara Kumar, 2018; Müller and Fritz, 2015; Zheng et al., 2016). However, a fewer number of electrodes did not yield reliable results, particularly Müller and Fritz (2015) achieved 71.36% accuracy, Zheng et al. (2016) achieved accuracy ranging from 50% to 62.5%, and Bairavi and Sundhara Kumar (2018) achieved accuracy ranging from 56% to 82%. It was found that the highest number of electrodes used in this review was 14 for consumer-grade EEG device (Blanco et al., 2019; Katsigiannis and Ramzan, 2018; Liu et al., 2018) and 62 for a research-grade device (Zheng et al., 2014) and Zheng et al. (2019). Accuracy results when using research-grade EEG devices are higher, while the results of Emotiv EPOC with 14 electrodes fluctuate from 61.84% to 92.26%. The results presented might also be affected by other factors, mostly related to the number and type of extracted features and the selected machine learning technique.

Products from the major suppliers of commercial EEG devices such as Emotiv, OpenBCI, and NeuroSky are readily accessible in the current market. Although these devices are identical in terms of affordability and portability, there are a few points that researchers might need to consider before using these devices. Table 4 shows a list of advantages and disadvantages of adopting these devices. NeuroSky products have the lowest price compared to other brands. They are simple to use, but having only one EEG

Table 4

Advantages and disadvantages of reviewed commercial EEG products.

Model	Advantages	Disadvantages
Neurosky MindWave Mobile 2	Low price Compact Wireless Practical	Only 1 sensor Bad quality for EEG experiments
Emotiv EPOC	14 sensors High signal quality Ready to use assembly Less artifacts Modern design	10–15 min setup time Non-dry, gel sensors, comfortability Requires a pro license SDK for raw data access
OpenBCI	Open-source Configuration options High customizability Available 3D print-ready headset models	Need to assemble yourself Requires significant pre-processing

electrode located at the forehead might lead to inconclusive results. According to our review, NeuroSky might not be that effective in emotion recognition studies. A study by Roesler et al. (2014) confirms this by claiming that the MindWave device is not suitable for ordinary tasks as the eye state classification (open or closed), having an accuracy of only 43.52%. Due to the availability of one sensor, higher-order EEG studies or control applications should not be feasible. However, NeuroSky products are a good option for applications such as tracking attention and meditation (Chen and Wang, 2018; Vasiljevic and de Miranda, 2019).

Based on the accuracy results presented in Table 3, OpenBCI can obtain similar features as a research-grade EEG device used by Javaid et al. (2015) and Zheng et al. (2014). At certain combinations of features and machine learning techniques used, Emotiv EPOC+ and NeuroSky devices can also provide comparable results (Bairavi and Sundhara Kumar, 2018; Liu et al., 2018). In this review paper, several studies used the Emotiv EPOC, which has 14-channels (+2 reference sensors) that apply the saline solution on the human scalp to improve the EEG data quality and it comes in a ready-to-use assembly, unlike OpenBCI products. Having more channels helps explore more brain regions simultaneously, allowing the researchers to collect more widespread signals. Nevertheless, in a study by Grummett et al. (2015), scientific comparison of the Emotiv EPOC+ to a research-grade g.tec system ("g.HIAMP | g.tec medical engineering," n.d.) was conducted. The study results confirmed that the low-cost, wireless systems might be employed for EEG-based experimental studies. Speaking of OpenBCI, initially, it was intended for building BCI systems and high school students and do-it-yourself (DIY) neuro-hackers. It comes as a circuit-board, while other parts can be purchased separately or even 3D printed, allowing researchers to assemble the device based on their research objectives.

Moreover, open-source hardware, firmware, and GUI of OpenBCI products are publicly available and easily modified for special projects. Some other advantages of OpenBCI include compatibility with different electrodes, headset mobility, and up to 16 channels. All these make OpenBCI more preferable over other commercial EEG devices. In this review, studies that used the OpenBCI device achieved an accuracy of 87.62% when using the SVM classifier (Javaid et al., 2015) and 70% when using the KNN classifier (Lakhan et al., 2019). A review study by Sawangjai et al. (2019) suggested that OpenBCI with 16 electrodes and Emotiv EPOC+ with 14 electrodes are the best option for EEG-based emotion recognition. The accuracy results may vary depending on various factors, such as the number of participants involved, type of stimuli used, features extracted, and machine learning algorithms adopted. Future studies can focus on the best conditions to improve human

emotion recognition accuracy using low-cost consumer-grade EEG devices.

From Table 3, the following results can be deduced:

- Out of the ten presented studies, nine studies split emotional states into two levels (high/low), and only one study [Zheng et al. \(2016\)](#) split into three levels (high/moderate/low), having the lowest accuracy results of 50%. It is somewhat agreeable, as fewer levels are easier to be classified, leading to better accuracy. Some studies that split emotions into two levels got low results, which might be caused by various features and classifiers used or the number of electrodes utilized.
- Both [Katsigiannis and Ramzan \(2018\)](#) and [Liu et al. \(2018\)](#) have used Emotiv EPOC and SVM-RBF classifier with the same PSD features extracted, but they have different pre-processing methods. They also have a different number of participants and gender balance. [Liu et al. \(2018\)](#) obtained a significantly better accuracy than [Katsigiannis and Ramzan \(2018\)](#). It might be mainly caused by the classification strategy followed by [Liu et al. \(2018\)](#), which consists of three SVM-based classification layers. The first layer detects neutral and non-neutral emotions. The second layer distinguishes the previous layer's results into positive and negative emotions, and the last layer detect emotions as joy, tenderness, amusement, sadness, anger, disgust and fear.
- [Zheng et al. \(2014\)](#) used research-grade device and KNN, GELM and linear SVM-Linear as classifiers. In 2019, the same authors conducted another study with similar experimental settings but with more extracted features and more participants [Zheng et al. \(2019\)](#). Eventually, [Zheng et al. \(2019\)](#) obtained higher accuracy with GELM and relatively same accuracy with KNN and linear SVM-Linear algorithms. Although they had a different number of participants, the difference of 5.4% in GELM results is significant enough to be affected only by the number of participants. It is more reasonable to claim that it was caused by different features tested in [Zheng et al. \(2019\)](#), as more features give more information to classify different emotional states. Notably, [Zheng et al. \(2019\)](#) also tested SVM-RBF. It resulted in an accuracy reduction by 5.05% compared to SVM-Linear, and 5.87% compared to SVM-Linear from an earlier study in 2014. The reason might be that linear SVM is a parametric model, whereas SVM-RBF is not. The complexity of the SVM-RBF increases with the size of the training set.
- [Lakhan et al. \(2019\)](#) and [Katsigiannis and Ramzan \(2018\)](#) utilized the PSD features, but [Lakhan et al. \(2019\)](#) used OpenBCI and a trained K-Means Clustering classifier. [Katsigiannis and Ramzan \(2018\)](#) used Emotiv EPOC and SVM-RBF classifier. [Lakhan et al. \(2019\)](#) slightly outperformed [Katsigiannis and Ramzan \(2018\)](#). One interesting finding is that apart from having different pre-processing steps, [Lakhan et al. \(2019\)](#) only use 8 electrodes EEG device, whereas [Katsigiannis and Ramzan \(2018\)](#) use 14 electrodes. Notably, the number of participants in a study by [Lakhan et al. \(2019\)](#) is twice as much as in the experiment by [Katsigiannis and Ramzan \(2018\)](#). The difference in results can be caused by a different group of participants and gender balance.
- [Javaid et al. \(2015\)](#) confirmed that a higher number of electrodes yields high accuracy results. It was shown that using OpenBCI with SVM-RBF classifier and Higuchi's FD features yields the best accuracy of 68.39% with 3 electrodes and 87.62% when the number of electrodes increased to 8. Notably, the recruitment of participants in [Javaid et al. \(2015\)](#) has more males than females. This inconsistency might be skipped in the participant selection process.
- In a study by [Zheng et al. \(2016\)](#), the NeuroSky Mindwave device was used during the experimental study. The study focus on one emotion, anxiety. KNN and SVM-RBF algorithms were used. Accuracy rates of 62.5% and 50% were observed, respec-

tively. The number of electrodes might cause this, and 3 levels (high/moderate/low) of anxiety examined. However, in all other studies where the KNN algorithm was used, the accuracy results ranged from 56% to 70.43% ([Bairavi and Sundhara Kumar, 2018](#); [Blanco et al., 2019](#); [Zheng et al., 2014](#); [Zheng et al., 2019](#)).

- Studies by [Javaid et al. \(2015\)](#) and [Liu et al. \(2018\)](#) have shown the best accuracy rate among all the reviewed studies. 87.62%, and 86.43%, respectively. The reason for that might be the type of features extracted and the machine learning classifier used. Notably, [Javaid et al. \(2015\)](#) involved only 8 participants, while [Liu et al. \(2018\)](#) had 30 participants. According to these data, we can infer that Emotiv EPOC with 14 electrodes and OpenBCI with 8 electrodes are the best option for consumer-grade emotion recognition experiments.
- Past studies show they have tested the collected dataset in multiple classification algorithms. According to our review, it is unfair to highlight the best algorithm or best features extracted due to the large variety of machine learning techniques and the high number of parameters in each algorithm. Finding the best classification method can be challenging, including the differences in procedures and a wide choice of EEG features. Hence, we observed that researchers put the sample in more than one machine learning algorithm and find the most suitable algorithm to describe their experiment. Therefore, further investigations are required to find the best combination of these aspects for consumer-grade EEG products. It is recommended to consider as many algorithms as possible to identify the proposed model's reliability, including pre-processing and feature extraction stages.

4. Challenges and recommendations

For years, research-grade devices with wet electrodes have been employed, giving high-quality data to test different machine learning models ([Koelstra et al., 2012](#)). The setting time for wet electrodes and the high cost of research-grade devices make it difficult for beyond the lab applications. In this case, dry electrodes are preferred. However, it is an open question of whether dry electrodes can provide similar signal quality. It has been reported that dry electrodes are comfortable but contain more artifacts than wet electrodes ([Guger et al., 2012](#)). Lately, low-cost consumer-level EEG wearables with high portability and dry electrodes appearing in the market. Despite the attractive characteristics, these EEG devices still have some drawbacks, such as limited long-term signals. Continuous pressure is required to maintain contact between the skin and dry electrodes, which is likely to disturb or even hurt when using for a long time ([Norton et al., 2015](#)). Hence, an extensive exploration should be conducted to affirm the benefits of dry electrodes further. Advanced solutions for electrode materials, such as porous ceramic "semi-dry" electrodes ([Li et al., 2016](#)), or electrode location, such as in-ear placement ([Mirkovic et al., 2016](#)), were suggested. Nevertheless, their performance for emotion recognition to be verified.

Before using a consumer-grade EEG device, it is recommended to estimate device capacity with a standard testing manual based on study objectives. The best way to do it is by reviewing relevant literature that employed the same device, but the reviewed literature's quality and scope must be considered. Additionally, it is a good practice to collect EEG data using consumer-grade device and research-grade device for the same study framework to compare data quality and accuracy results ([Hu et al., 2019](#)). EEG signal recording is vulnerable to outside artifacts even in a shielded room, not to mention the real-life environment. Thus, consumer-grade devices with a relatively low number of electrodes might have poor signal quality. Some artifact removal algorithms, such as ICA, strongly depend on spatial information, whereas a low number

of electrodes would highly impact their performance. Despite the availability of ways to process single-channel signals, such as empirical mode decomposition (Mijović et al., 2010), a more powerful pre-processing method for signals recorded with consumer-grade EEG devices is still required.

Numerous machine learning algorithms have been tested and verified to classify emotional states, including Naive Bayes, Random Forest, SVM, K-Means Clustering, KNN and GELM. Lately, neural network algorithms are getting popular for EEG-based emotion recognition, including deep belief networks (DBN) and convolutional neural networks (CNN) (Chao et al., 2018; Yang et al., 2018). Accuracy results vary from 70% to 90% for recognizing two or more discrete emotional states. There is no sufficient evidence to declare the optimum machine learning algorithm for emotion recognition tasks, especially when using the consumer-grade device. The small number of electrodes, lightweight and neat design of consumer-grade EEG devices make them more user-friendly, thus have great potential for applications in a real-life environment, such as education, entertainment, and medicine. There is a need to identify and standardize the general procedure for emotion recognition studies when using a consumer-grade EEG headset.

Some previous studies utilized self-created emotional stimuli (Lakhan et al., 2019; Zheng et al., 2016). Some studies proposed models assuming a stationary emotional state for a relatively long period, generally more than 10-second epochs (Katsigiannis and Ramzan, 2018; Koelstra et al., 2012). The stimuli presented to participants might encounter some inconsistencies in terms of duration, as emotions are dynamic and can rapidly change depending on time and the type of stimuli (Trull et al., 2015). Thus, to avoid these emotional fluctuations and collect the participant's correct emotional response, the stimuli' duration is recommended to be in the range of 5–10 s. The reason for selecting this range is to provide adequate time for inspecting the stimuli to get involved in it so that the participant's emotional state matches the targeted emotions. At the same time, this range would reduce the chances of emotional fluctuations. Building an automatic labeling system that can precisely reflect the dynamic emotional experience based on data from other modalities, such as facial expressions, peripheral physiological data, video content analysis, might be very useful for future research.

In daily human life, emotions occur when someone is passively observing stimuli (e.g., watching funny videos on YouTube) and actively involved in emotional interaction (e.g., telling a funny story to friends). These two types of events are frequently co-occurring and interact in a complex manner. However, previous studies rarely involve both of these events. Researchers might focus on linking two emotion elicitation techniques to promote emotion recognition systems' compatibility with passively and actively induced emotions. It would be required to test the proposed model using one elicitation method on the other method's data. Additionally, it is recommended to train a new model including both passive and active elicitation stimuli data. It is a good practice to verify emotion recognition models with different data types, which will help researchers estimate these models' application boundaries. A well-controlled experimental setup would facilitate signal acquisition and analysis but possibly bound the generalization of the data. Hence, researchers should focus on identifying the balance between these two paradigms.

There are various publicly available benchmark EEG datasets, such as SEED, MAHNOB-HCI and DEAP. (Katsigiannis and Ramzan, 2018; Koelstra et al., 2012; Liu et al., 2018). Generally, the benchmark dataset provides a relatively low sample size, with 30–60 participants involved and 20–40 emotional stimuli. Insufficient sample size can hinder model performance, mainly when dealing with advanced neural network models. Therefore, some studies try to build their own dataset (Lakhan et al., 2019; Liu and Sourina, 2014). Since the collection of large-scale EEG data is not a trivial task,

it is strongly recommended to conduct it cooperatively with selected stimuli and collection protocol followed across multiple institutions, as was done in some other areas (Brazma et al., 2001; Deus et al., 2012). It is also good to record data from other peripheral physiological modalities, preferably simultaneously with EEG signal acquisition. The combination of information from various modalities, such as facial expression, body movement, eye blinks, and heart rate, might help analyze emotional states and improve final model performance (Soleymani et al., 2012).

5. Conclusion

The recent evolution of EEG technology and the availability of consumer-grade products in the current market have promoted emotion recognition studies and novel experiments to identify better strategies. These studies' outcomes can be beneficial in various fields, such as medicine, education, entertainment, and marketing. This paper systematically reviewed ten studies that utilized popular consumer-grade EEG devices with attention to the number of participants involved, the type of stimuli used, features extracted, and machine learning algorithms adopted. The findings are aimed to support future investigations by young practitioners and researchers who are new to this field.

EEG-based emotion recognition comprises multiple factors that need to be accounted for in research methodology. It was observed that the general procedure of EEG-based emotion recognition starts with the proper recruitment of participants and the selection of stimuli, followed by data acquisition, data pre-processing, feature extraction, and emotion classification. The reviewed studies' results indicate that participants' gender and age might be the factors that affect the experiments' outcomes. Besides, researchers should also consider the usability of EEG devices, enhance computational methods, and combine various modalities in the core of their studies. The study also showed no particular feature extraction technique or classification method called an exclusively best option for all cases. The strategy relies on the concrete system paradigm and objectives. An implication of this is the possibility of considering as many machine learning algorithms as possible to define the proposed model's reliability, including pre-processing and feature selection stages. Then, compare the results with various features and techniques before choosing the best model that satisfies the given objectives. Finally, we found that low-cost consumer-grade EEG devices can equally perform as research-grade devices.

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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